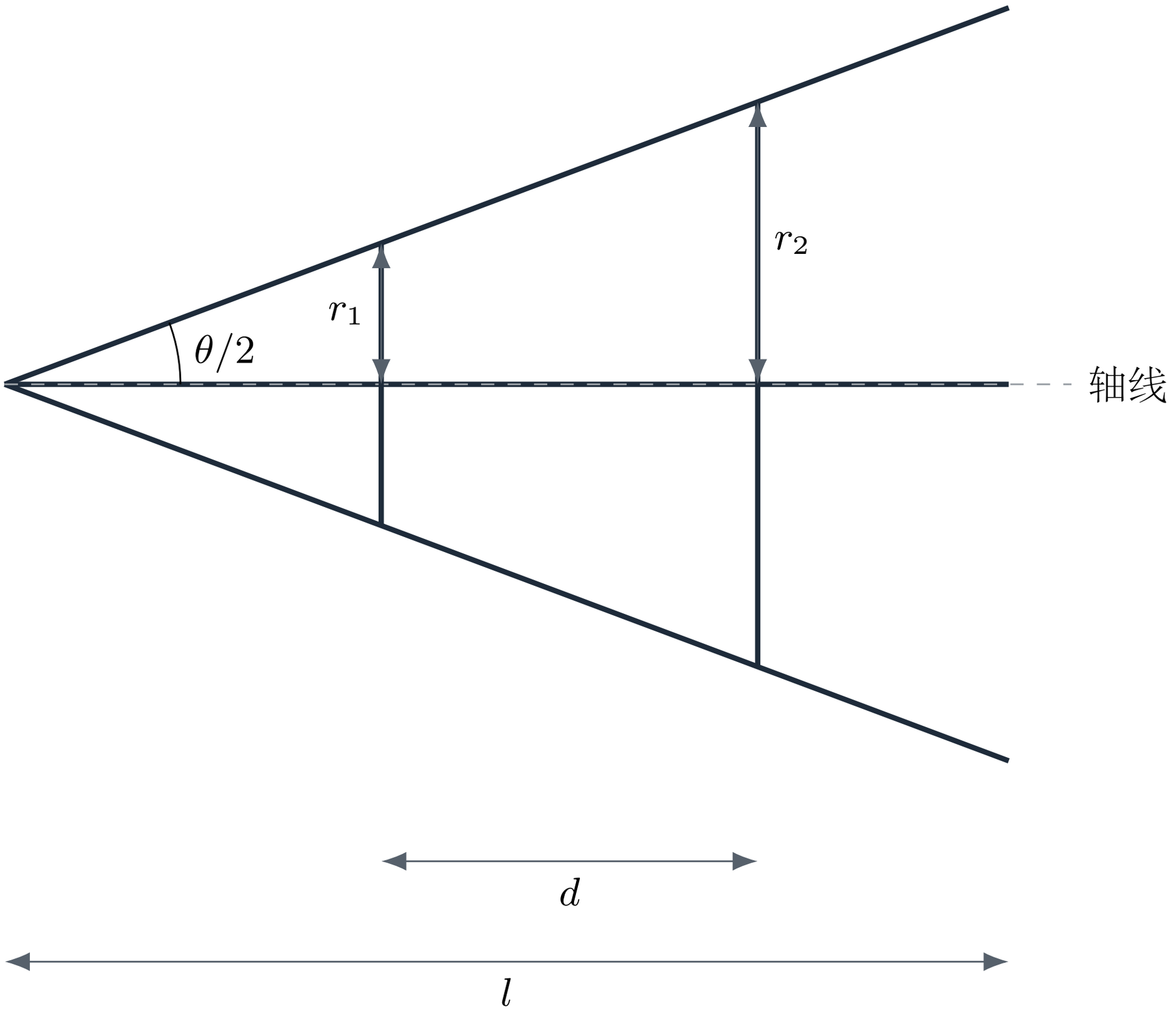
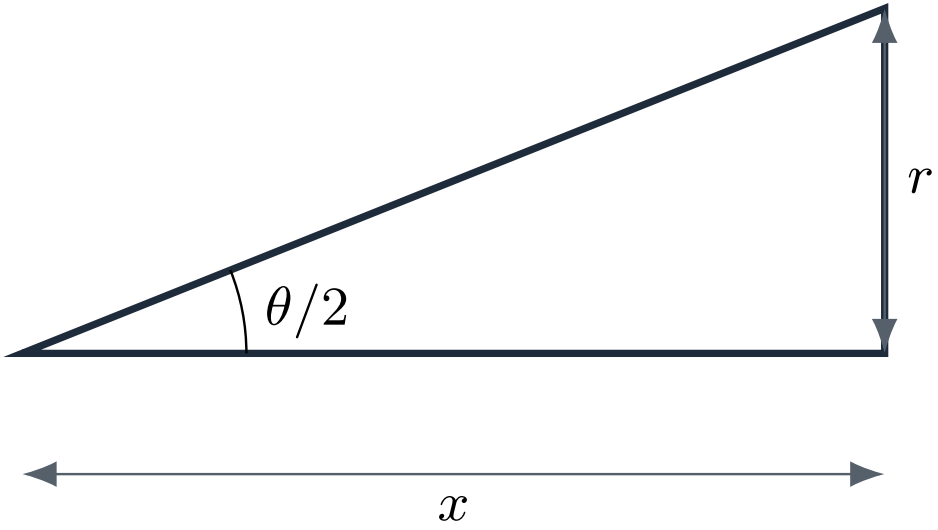
轴截面；横向移位改变两处半径





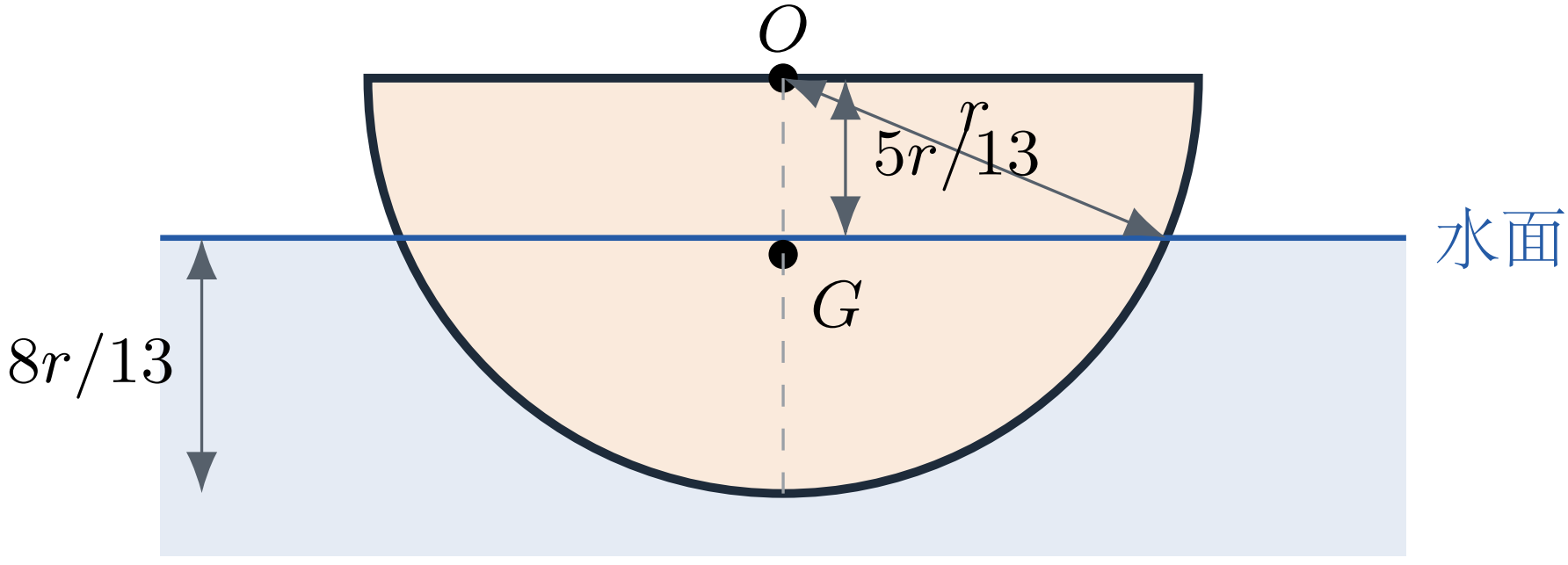
$$r = x\tan(\theta/2)$$

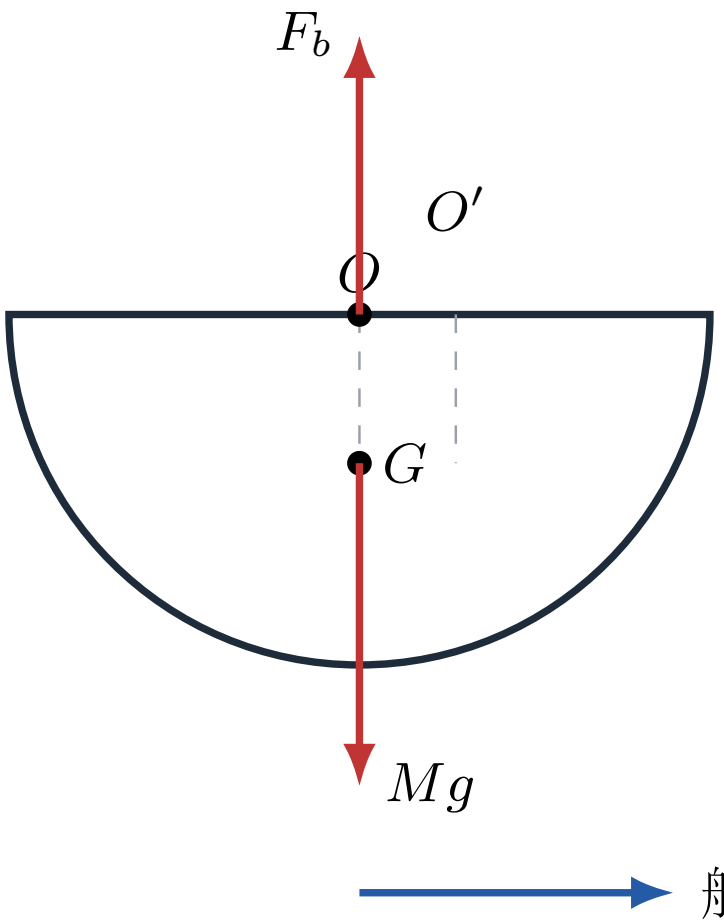
$$r_1 + r_2 = \text{const}$$

$$\omega_1 r_1 = \omega_2 r_2$$

长带近似与精确切线长度分开使用

静态截面：水线穿过半圆的 5—12—13 弦





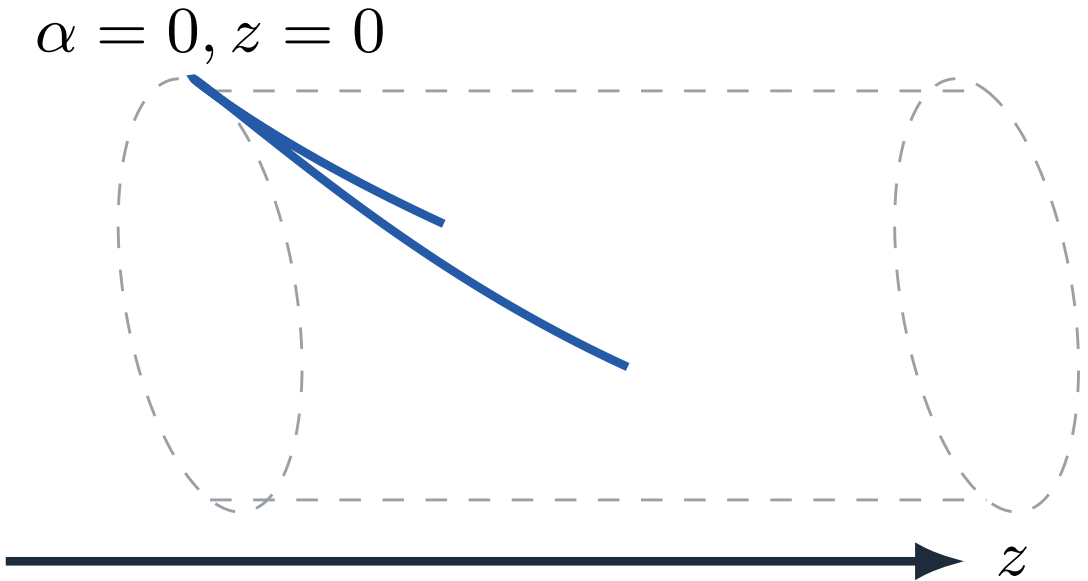
O 非固定铰链；G 的横坐标近似固定

$$I_G = I_O - Mh^2$$

$$T_2 = 2\pi\sqrt{I_G/(Mgh)}$$

$$\Delta\theta_{\text{cycle}} = 4mrI_O/(MhI_G)$$

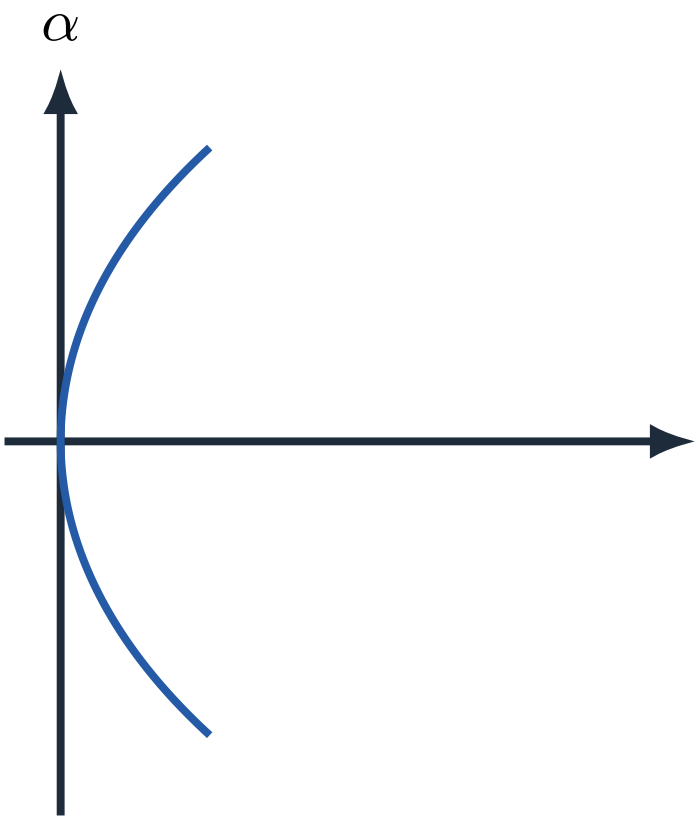
快跑既改变平衡角，也产生姿态跳变



两端 $\alpha = \pm\pi/2$

轴向坐标相同

$z = (r/\mu)(\cosh \mu\alpha - 1)$

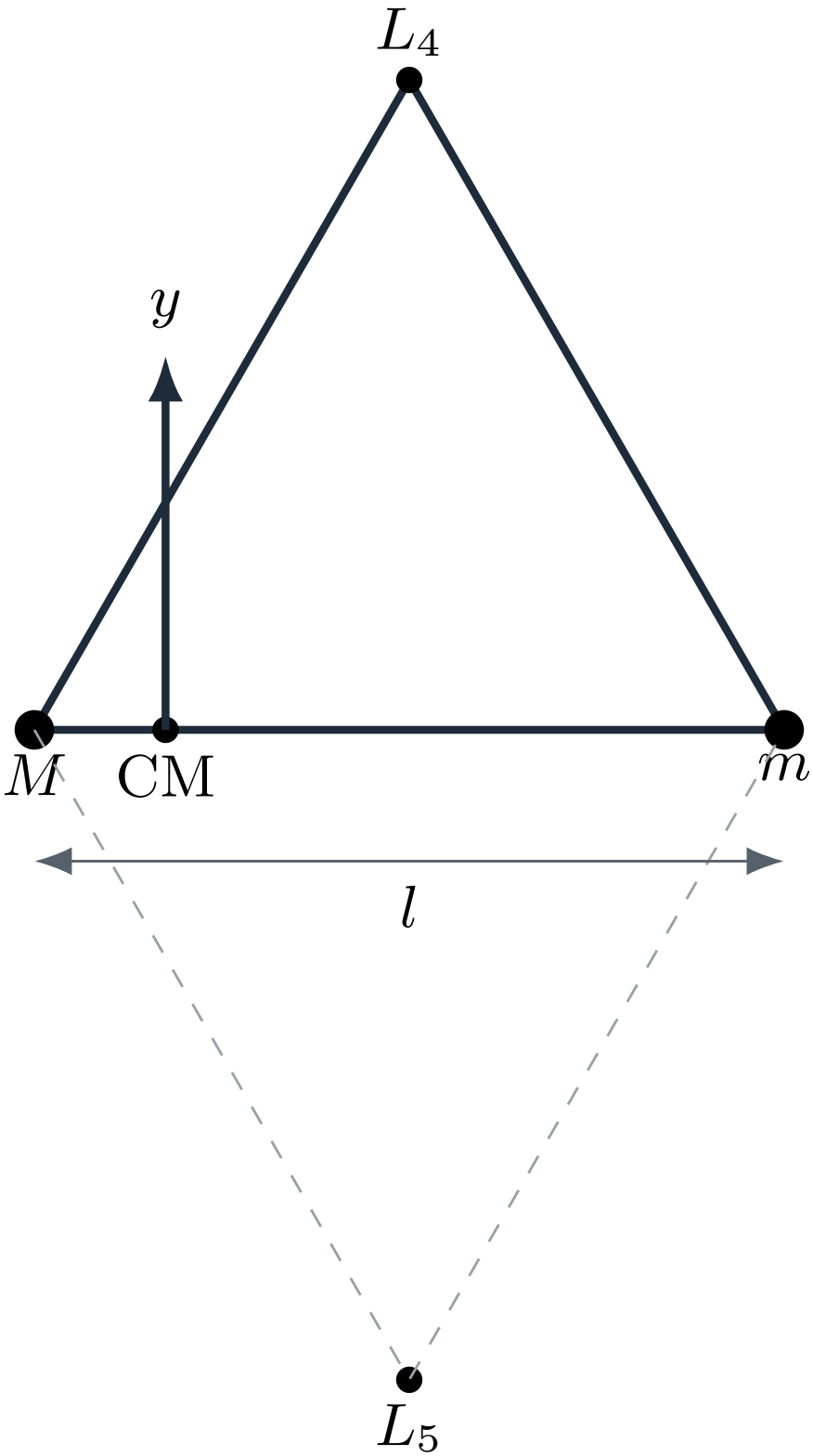


$$\rho = r$$

$$z/r = (\cosh \mu\alpha - 1)/\mu$$

$$-\pi/2 \leq \alpha \leq \pi/2$$

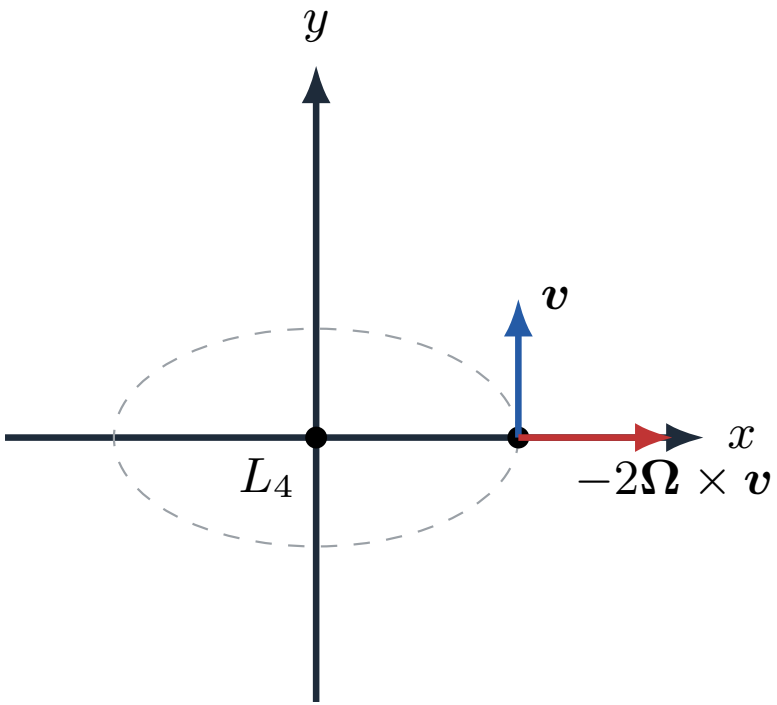
在 $\alpha = 0$ 处有竖直切线；关于横轴对称



三条边均为 l

$q = m/(M+m)$

角速度 $\Omega^2 = G(M+m)/l^3$

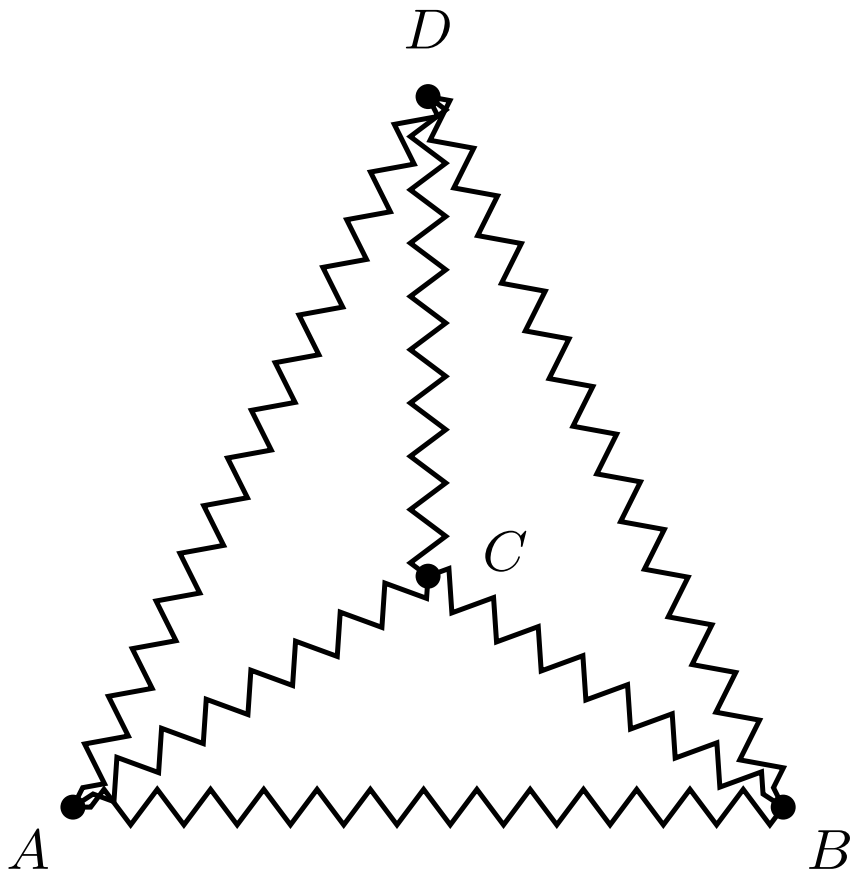


$$\lambda^4+\Omega^2\lambda^2+\tfrac{27}{4}\Omega^4 q(1-q)=0$$

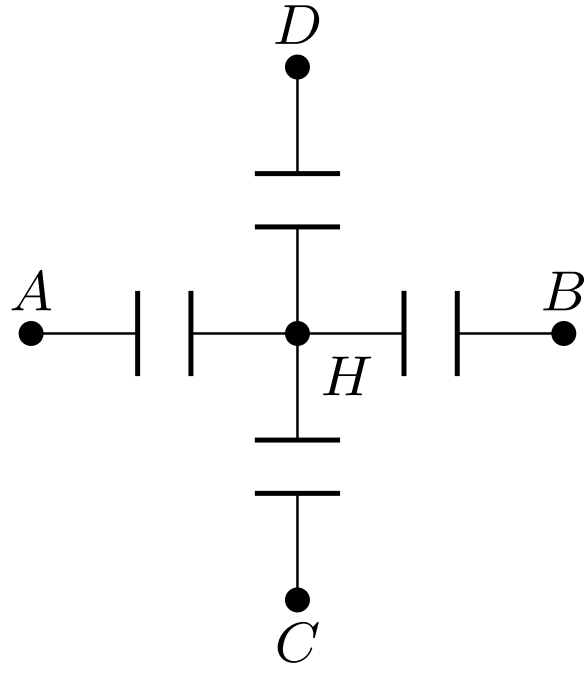
线性有界条件：$0<q(1-q)<1/27$

临界等号存在 $te^{i\nu t}$ 项

虚线为局部扰动示意，不代表一般闭合轨道



电阻子图：完整 K_4，交叉处不另设节点

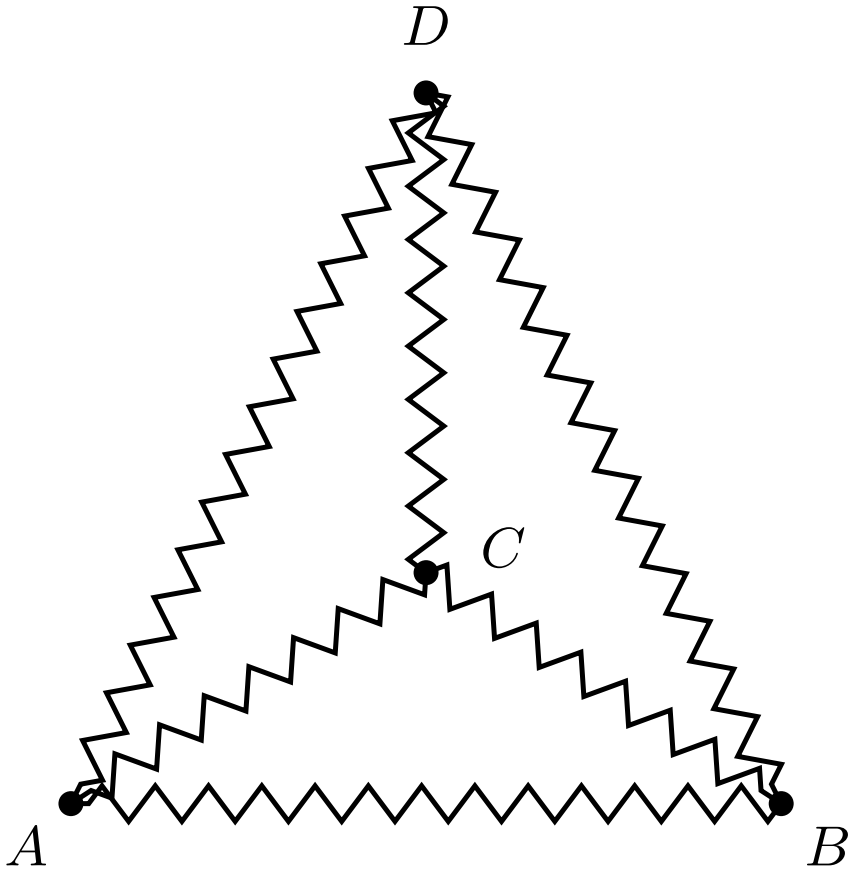


电容子图：四条 C 支路

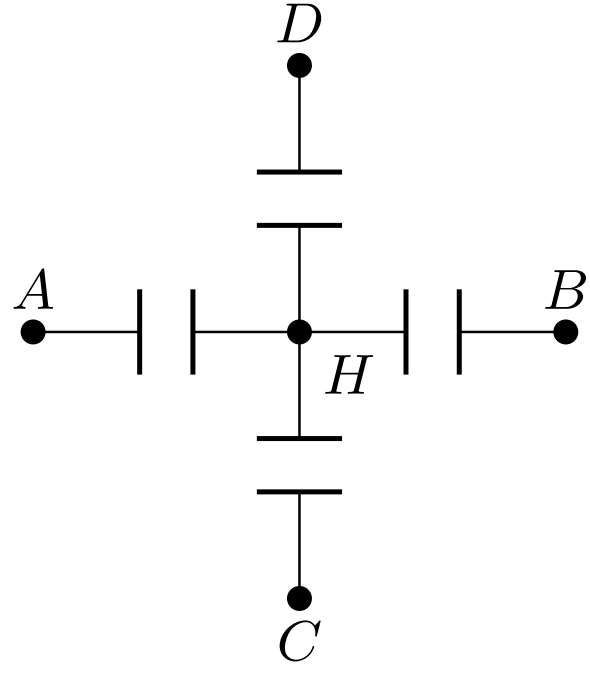
坐标原点为 H；$+z$ 指向 D

$+x$ 指向底面顶点 A，B 在 $+y$ 一侧

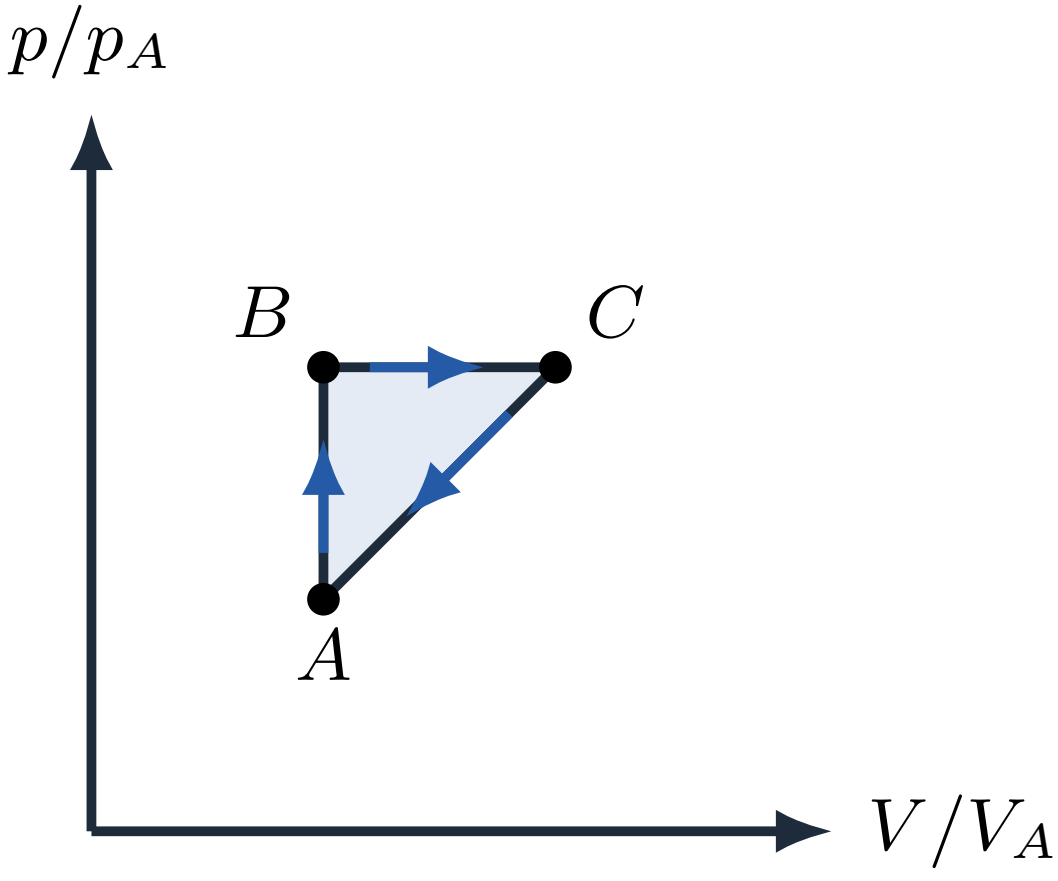
$$\sum_{j\neq i}\mathcal{E}_{ij}=0 \quad\Longrightarrow\quad V_i-V_H=0$$



电阻子图：完整 K_4，交叉处不另设节点



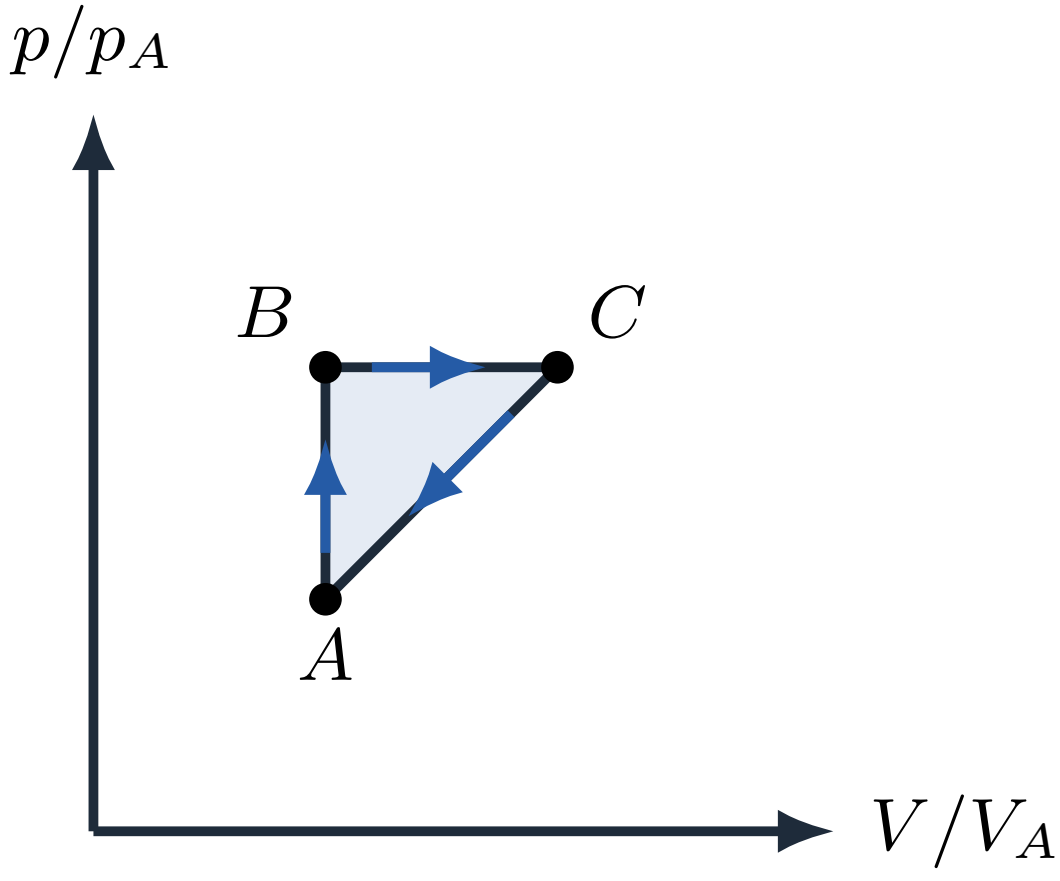
电容子图：四条 C 支路



$A \to B$: 等容

$B \to C$: 等压

$C \to A$: $p = p_A V / V_A$

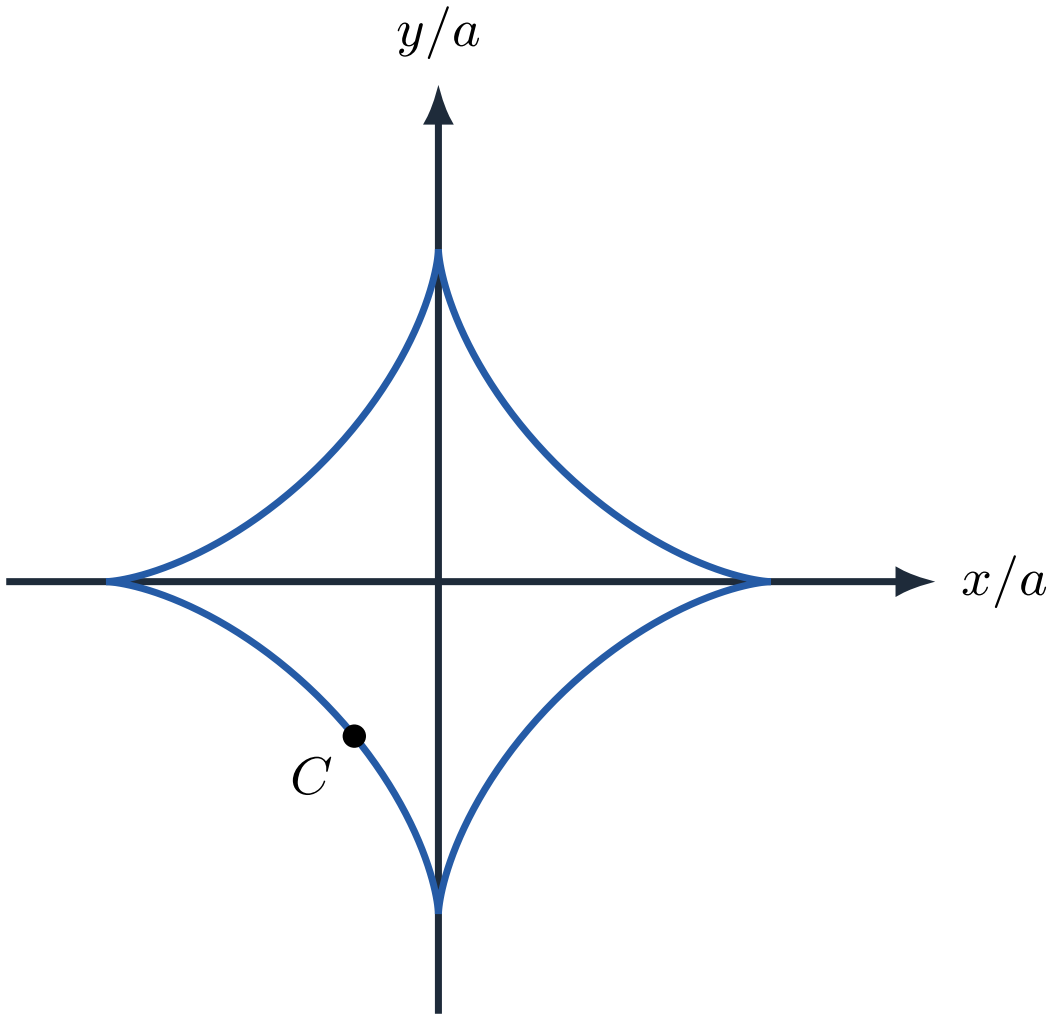


$A \to B$：等容

$B \to C$：等压

$C \to A$：$p = p_A V/V_A$

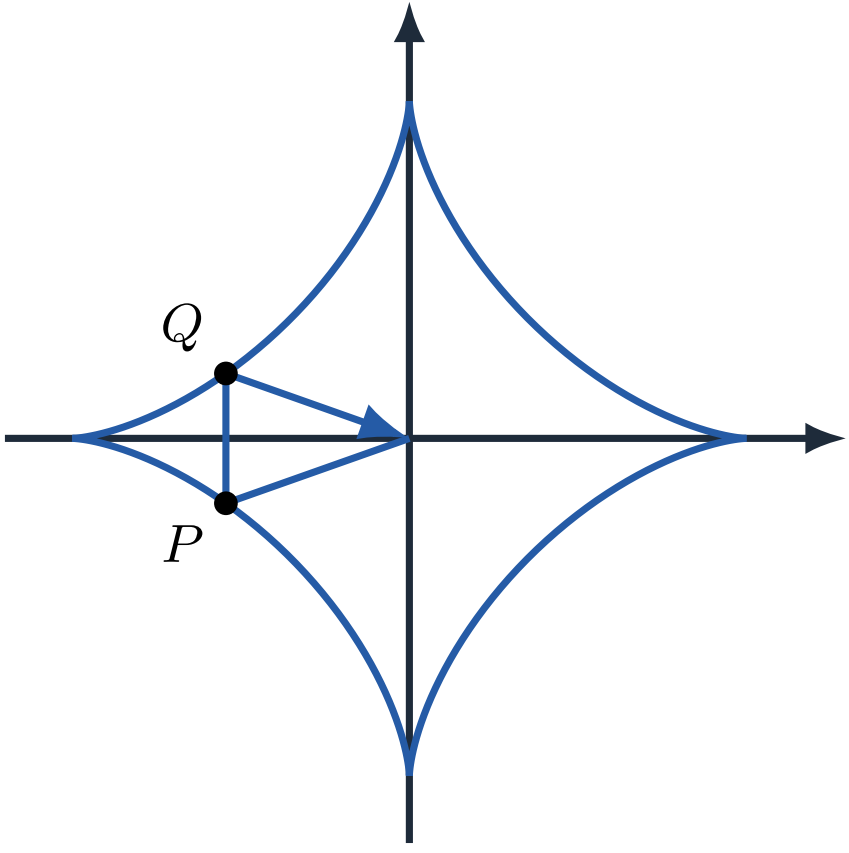
循环功 $W = p_A V_A/2$



$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$A = (-1, 0), \quad B = (0, -1)$$

图中 C 仅示意脱离点，坐标待求



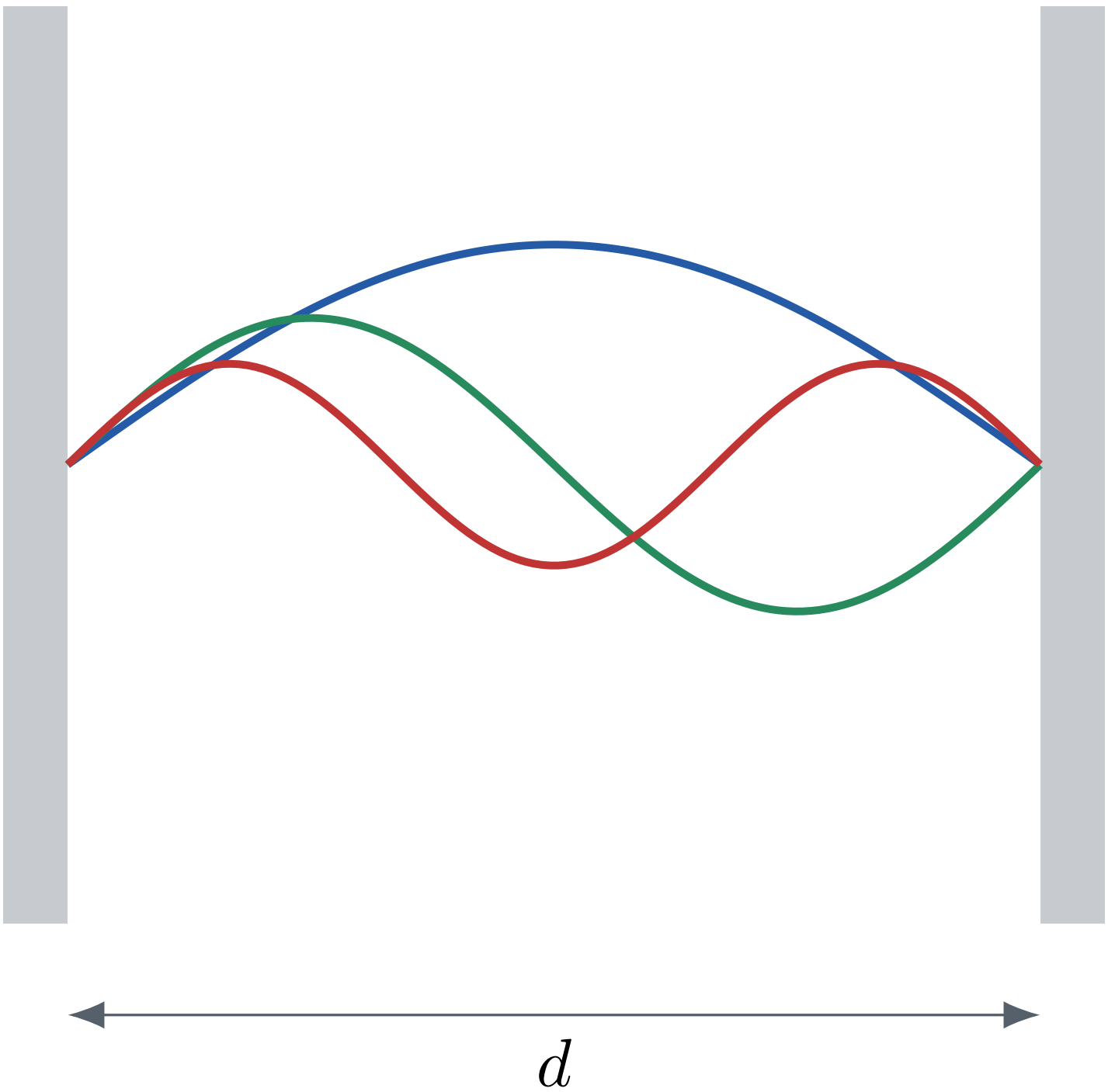
一条两次反射回到 O 的光路

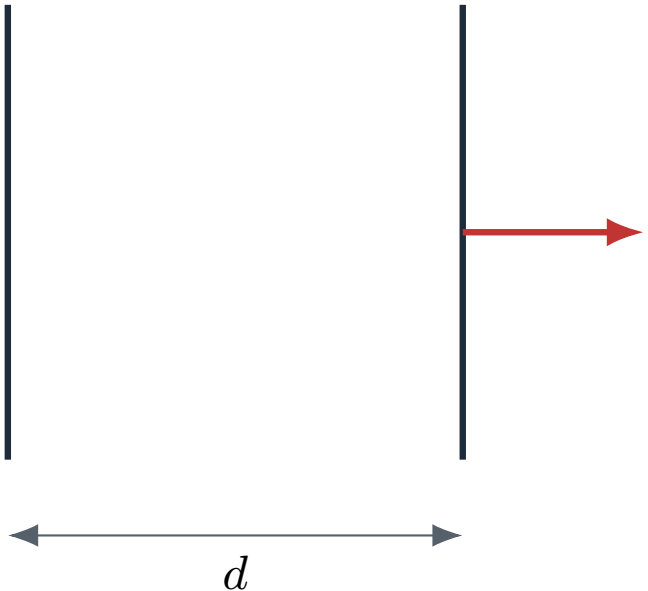
其镜像光路给第三个方向

角平分线方向为 $225°$

三个方向：$199.47°, 225°, 250.53°$

两平行板间的驻波模





$\nu_n = nc/(2d)$，$n < p$ 只取 $1, \ldots, p-1$

$E(d) = hc\,p(p-1)/(8d)$

固定这段的模数：$F_d = -E'(d)$

图示是有限截断模型的斥力；不等同实际 Casimir 力